

Progress Presentation-III

Automated Mobile Robotic Arm for Cold Storage Operations



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Project Guide

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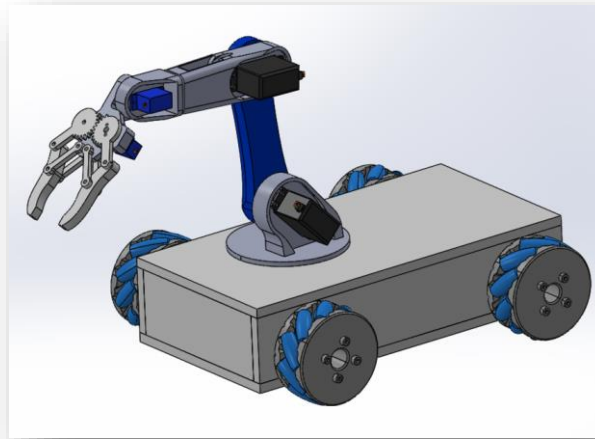
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Literature review



Relevant Technologies:

- Robotics and automation are already widely used in industries like logistics and cold storage. However, most of these systems are either stationary or have limited mobility. Technologies like automated conveyor belts, robotic forklifts, and stationary robotic arms are commonly used to improve efficiency in cold environments.

Gaps in Knowledge:

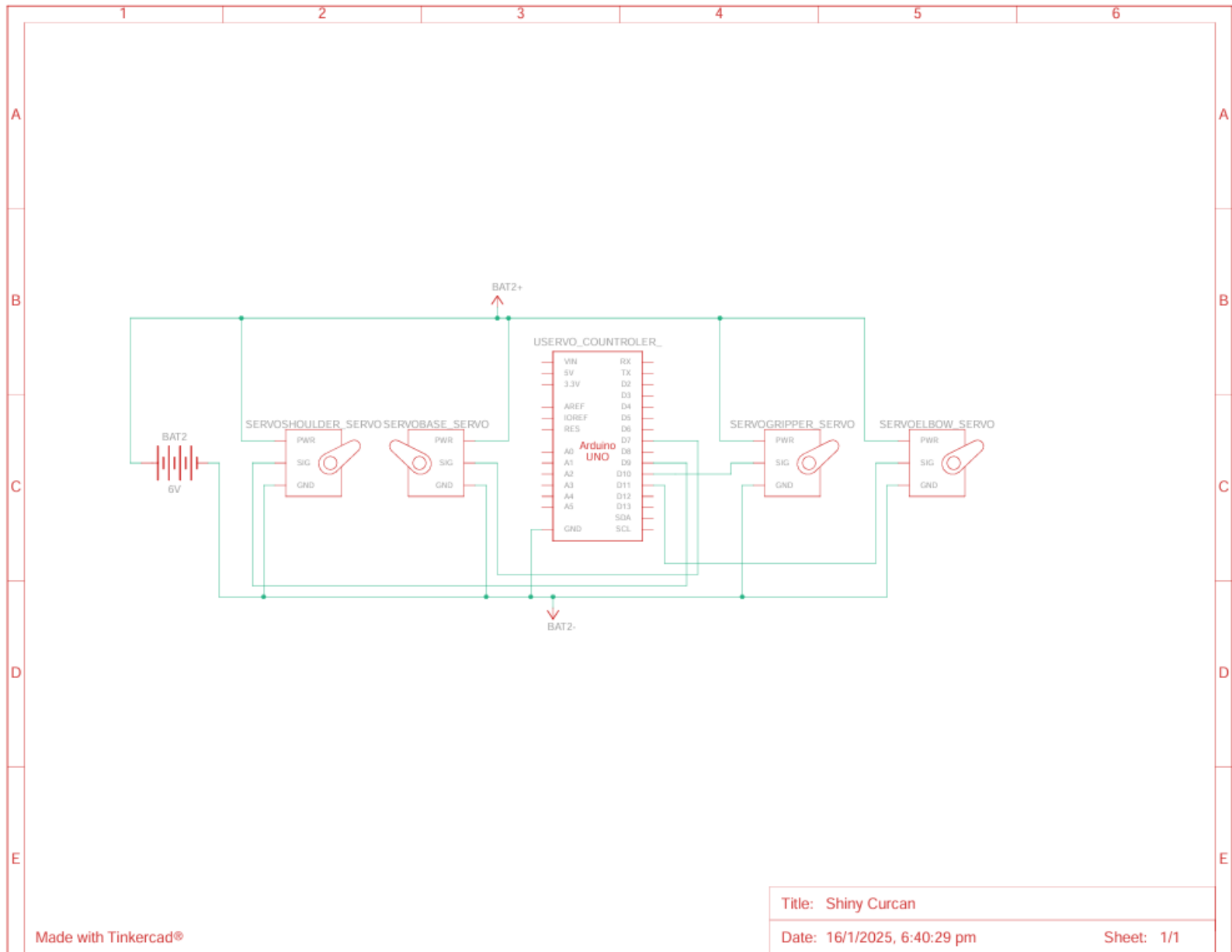
- Although many studies focus on robots working in challenging environments, there is limited research on mobile robotic systems specifically designed to operate in cold storage. Most robotic systems are not built to handle the combination of mobility, precision, and durability required in such low-temperature environments.

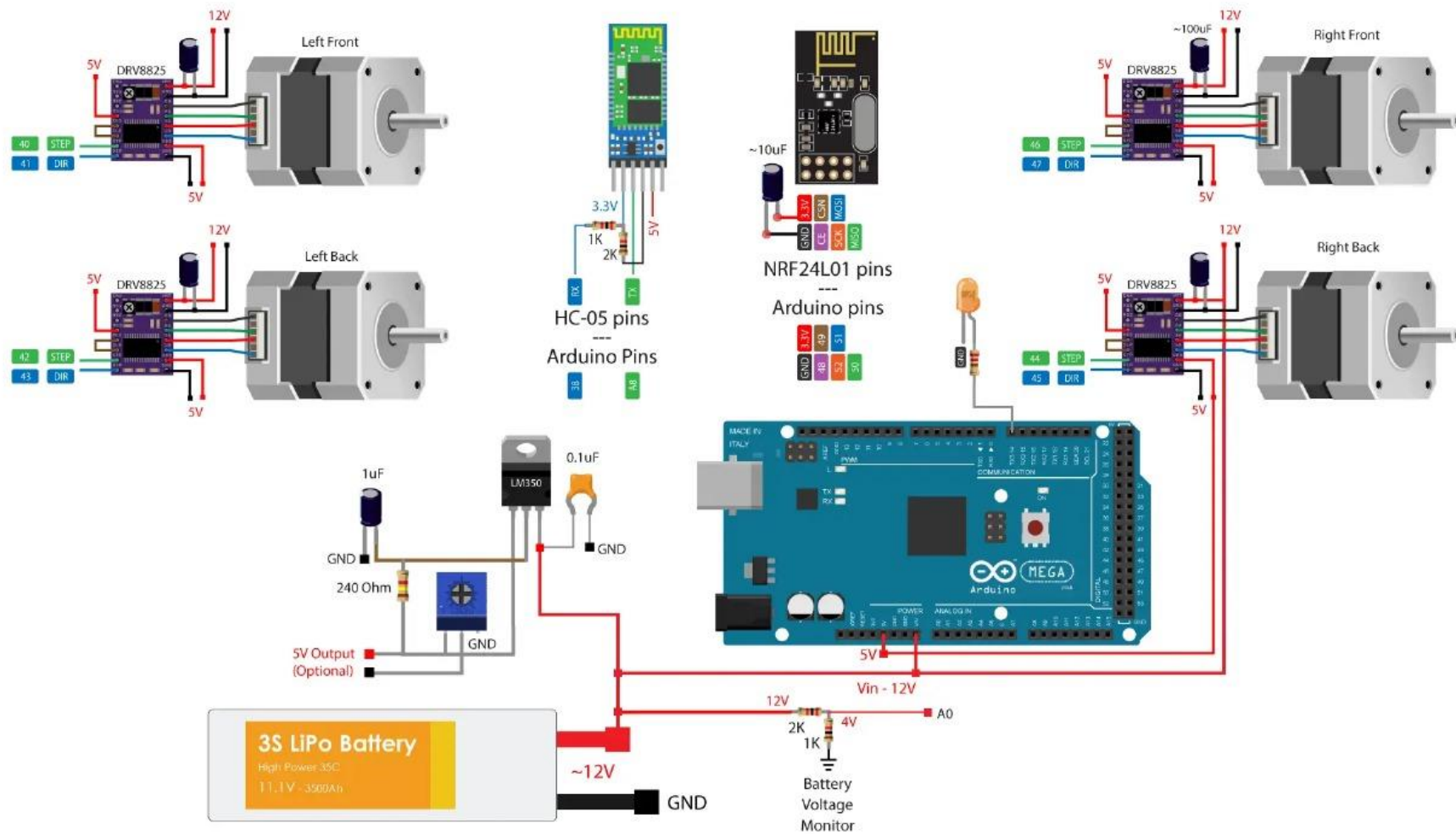
Project Setup

Components:

- Servo Motors
- Servo Controller
- Battery
- Stepper Motor
- DRV8825 Stepper Motor Driver Module (High Current)
- Controllertech HC-05 master-slave 6pin Bluetooth Transceiver Module







Progress of project work

- **Accomplished Work:**
- Bought and tested components like MG995 servo motors, a servo controller, a Samsung Li-Ion battery, and an SG90 micro servo.
- Successfully ran the motors using software and Bluetooth control.
- Planning to start 3D printing parts.

FileFunctionsHelp

Servo 1

Servo 2

Servo 3

Servo 4

Servo 5

Servo 6

Servo 7

Servo 8

Servo 9

Servo 10

Servo 11

Servo 12

Servo 13

Servo 14

Servo 15

Servo 16

Servo 17

Servo 18

1482

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Open File

Save File

Reset Labels

Settings

Show Groups

Home All

Connection Type

USB

Bluetooth

Select Port Manually

COM6

Disconnect

Connected to COM6

Save Configuration to Controller

Generate Arduino Code

Function Control

F5

↑1

F7

←3

→4

F6

↓2

F8

Servo Sequencer

Delay

100

Go to Line

1

Speed

1

Beep

Home

Pause

Function

1 - Ctrl+Up

Comments for function

Add Move

Remove Line

Clear All

Default Delay

300

Start From Current Line

Loop

STOP

Run Sequence

Servo Idle

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

Clear

Future Work

- 3D printing the robotic arm

Next Steps

- Assembly Process



STEP 1 Put in position a 946 servo with the driving shaft forward



STEP 2 Fix the servo to the main base using the selftapping screws supplied with it



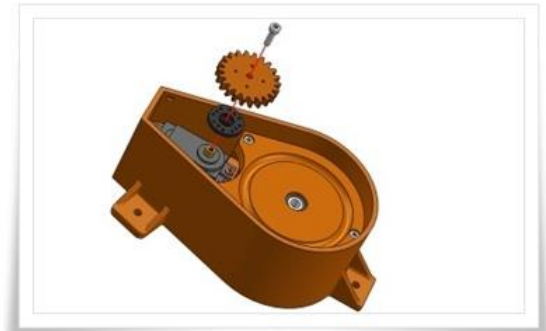
STEP 3 Insert 3 M3 nuts in the receptacles of the main base as shown.



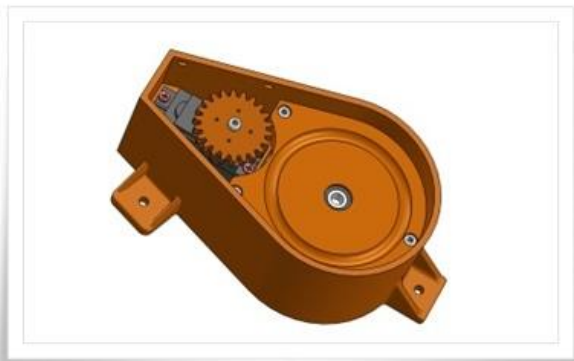
STEP 4 Insert the 606 bearings in its housing and attach the plate to the main base using 3 M3 screws.



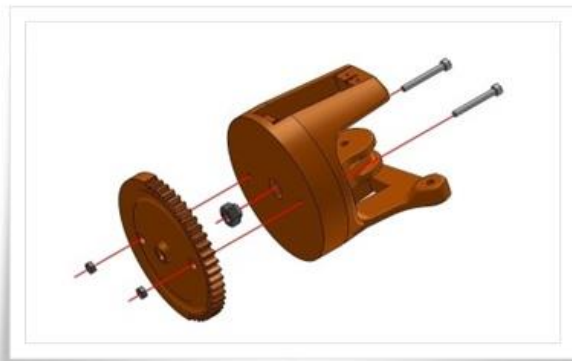
STEP 5 Verify the freedom of movement of the bearing.



STEP 6 Position the drive plate on the splined shaft and upper the driving printed gear. Add one or two small selftapping screw to connect plate and the gear



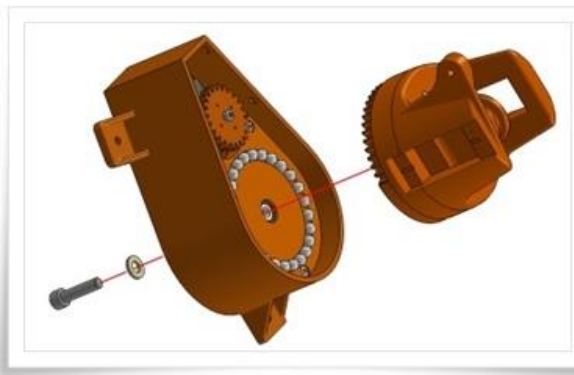
STEP 7 There are two driving gears available one has 22 theet and the other 25 I made two because during printing of the base I've got some deformation and the two axis distance became smaller



STEP 8 Insert an M6 self locking nut in the receptacle of the swivel base then place in position le geared base and fix it using a couple of M3 screws and nuts as shown.



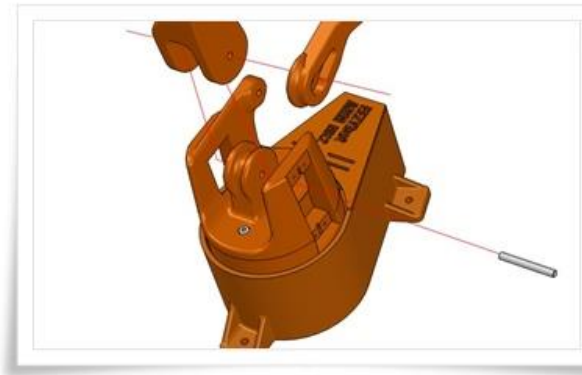
STEP 9 Fill the path using about 25 spheres with a diameter of 6mm.



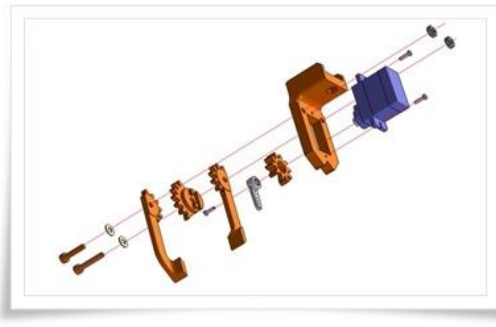
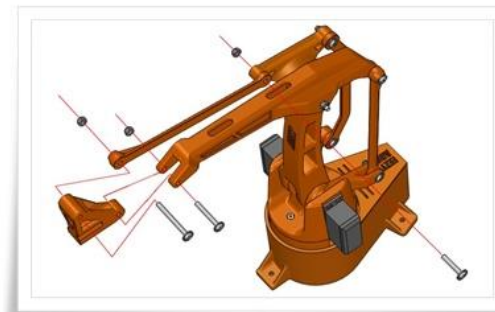
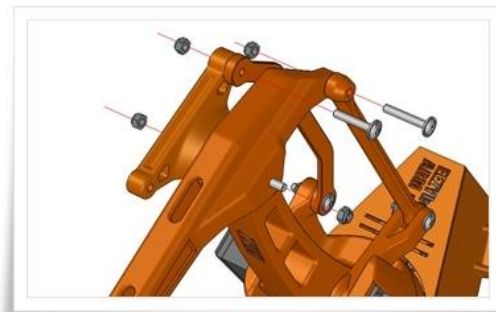
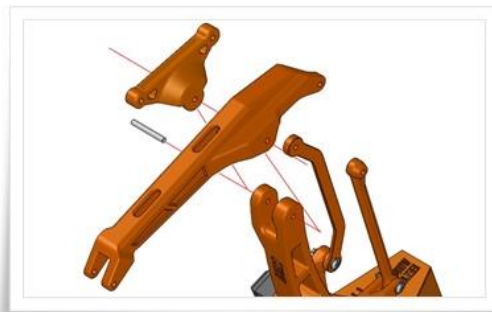
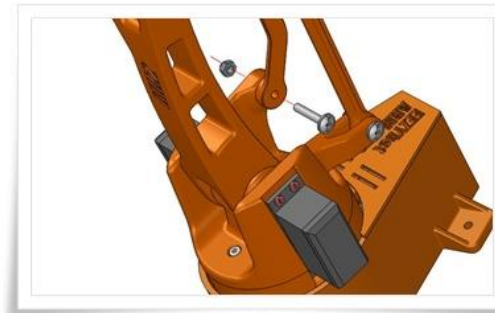
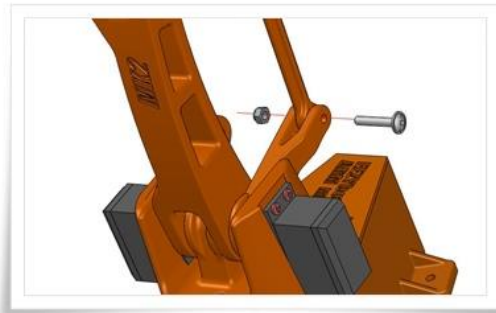
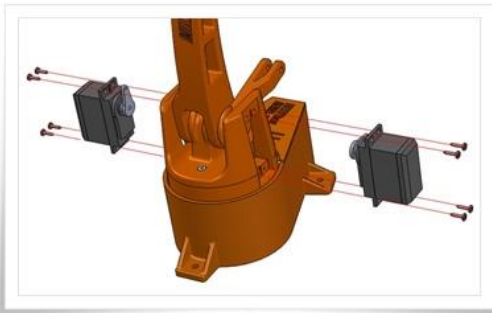
STEP 10 Keeping the main base flat and the swivel element in contact with it, connect the two elements using an M6 screw.



STEP 11 Now the main base is finished.

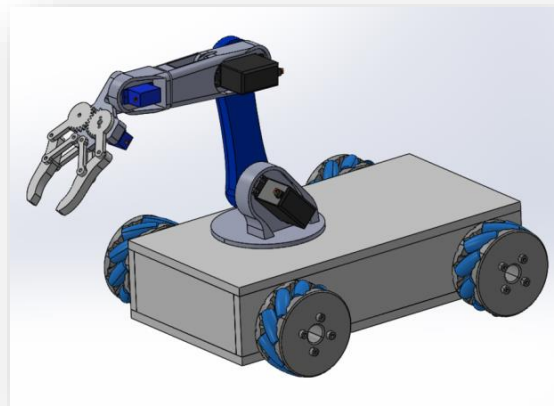


STEP 12 Put in position the main arm and the vertical drive lever, connect them with the main base horizontal axis using a 4mm dia rod 33 mm long.





now mechanically assembled, ready to be driven by the electronics.



Future Work

- Next Steps:
 - - Integrate the arm with the mobile platform using Mecanum wheels.
 - - Prepare the system for cold storage simulation and testing.
 - - Test for battery life, motor performance, and stability in low temperatures.

Conclusion

- **Summarized Progress:**
 - - Components tested and running successfully.
 - - 3D printing and assembly in progress.
- **Next Steps:**
 - - Finalize assembly, integration, and testing in a cold storage environment.

THANK YOU